

THE 1000 Hz BARE-METAL MCU TIMING TICK VS. LINUX MPU JITTER

Deterministic 175 µs safety reflex execution vs. non-deterministic 240 ms Linux tail latency blowouts

✓ DETERMINISTIC REAL-TIME REFLEX BRAIN (ARM CORTEX-M4 / FREERTOS · 1000 Hz TICK = 1000 µs)

IPC 15 µs	EKF / SE(3) 35 µs	QP CBF Solve 65 µs	d_stop Audit 40 µs	PWM 20 µs
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DETERMINISTIC SLACK MARGIN: 825 µs (82.5% Headroom)
Low-power sleep / background hardware diagnostic checks

TOTAL EXECUTION: 175 µs (P99 ≤ 178 µs – Zero Jitter!)

- **Zero Dynamic Allocation:** Static memory arrays, zero OS heap, zero garbage collection pauses.
- **Guaranteed Deadline:** Never misses a 1000 Hz PWM motor commutation tick under any physical condition.

✗ NON-DETERMINISTIC LINUX MPU TAIL LATENCY BLOWOUT (UNTRUSTED HOST BRAIN)

P50 = 22 ms Nominal Inference	P99 = 140–240 ms ⚠ GC Stall + UMA Bus Contention + Thermal Throttle	240x DEADLINE VIOLATION (240 ms vs 1 ms World Deadline) If Linux held motor authority, the robot would crash during every memory contention spike!
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- **Root Causes of Linux Jitter:** Python GIL locks, dynamic memory page faults, kernel context switches, and thermal throttling.
- **Why the Dual-Brain Architecture is Inescapable:** The real-time MCU absorbs all MPU jitter, keeping the robot stable and safe!